SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – COMPARATIVE ANALYSIS

Course Code:	CSA12	**Course Title:**	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	**Assessment:**	Comparative Analysis
Weightage:	30%	**Total Marks:**	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Section / Batch:	**B.TECH IT**	**Date:**	**20/08/2026**

Instructions to Students

- Identify the relevant concepts of L1, L2, and L3 cache, SRAM, DRAM, NAND Flash, MRAM, RRAM, virtual memory, paging, latency, bandwidth, and throughput.
- Analyze the memory requirements of different computer systems by considering cache levels, memory technologies, access latency, bandwidth, throughput, capacity, and energy consumption.
- Evaluate the performance characteristics of different memory technologies and determine their suitability for cache memory, main memory, secondary storage, and future hierarchical memory systems.
- Compute relevant performance measures such as access time, latency, bandwidth, throughput, hit/miss rate, average memory access time (AMAT), speedup, and energy efficiency wherever applicable.
- Interpret the results by assessing memory bottlenecks, performance improvements, technology trade-offs, and the suitability of each memory technology or memory hierarchy for a given application.

QUESTIONS

1. Evaluate L1, L2, and L3 cache levels based on access latency, bandwidth, and throughput. Analyze how each cache level influences processor performance and memory access efficiency.
2. Assess the performance characteristics of SRAM and DRAM with respect to access latency, bandwidth, and throughput. Justify the use of SRAM in cache memory and DRAM in main memory based on their respective characteristics.
3. Examine virtual memory, paging, and cache memory mechanisms in terms of access latency, memory utilization, and system throughput. Analyze how these mechanisms contribute to efficient memory management and overall system performance.
4. Analyze the performance characteristics of NAND Flash and DRAM with respect to access latency, bandwidth, and throughput. Evaluate their suitability for different levels of a hierarchical memory system and justify their applications.
5. Evaluate MRAM and RRAM technologies based on access time, throughput, energy efficiency, and endurance. Analyze their potential applications in future hierarchical memory systems and assess their performance advantages and limitations.

Code of Conduct

*I **<u>ABDUL WAHID. H / 192321079</u>** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ABDUL WAHID. H

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	**Level 4 – Exemplary (5 Marks)**	**Level 3 – Proficient (4 Marks)**	**Level 2 – Developing (2–3 Marks)**	**Level 1 – Beginning (0–1 Mark)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Marks Evaluation:

Q. No	Scope of Items **(20)**	Comparison Depth **(25)**	Use of Evidence **(20)**	Critical Thinking **(20)**	Clarity of Presentation **(15)**	**Total (out of 100)**
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	**/ 25**	**/ 20**	**/ 20**	**/ 15**	**/ 100**

Course Coordinator **Faculty Incharge**

Question 1: L1, L2, and L3 Cache — Latency, Bandwidth, Throughput

Modern processors use a multi-level cache hierarchy (L1, L2, L3) placed between the CPU core and main memory to bridge the large speed gap between the processor and DRAM. Each level trades capacity for speed: as we move from L1 to L3, access latency and capacity increase while per-access speed and bandwidth decrease.

Comparative Table

Parameter	L1 Cache	L2 Cache	L3 Cache
Typical Size	16–64 KB (per core, split I/D)	256 KB – 1 MB (per core)	2 – 64 MB (shared across cores)
Access Latency	1–4 CPU cycles (~0.5–1 ns)	8–15 cycles (~3–5 ns)	25–40 cycles (~10–20 ns)
Bandwidth	Very high (multiple bytes/cycle, closest to core)	High, lower than L1	Moderate, shared among cores so contention reduces effective bandwidth
Throughput	Highest — services almost every load/store	High for L1 misses	Lower per-core but sustains overall system throughput by avoiding DRAM access
Placement	On-core, private	On-core or on-die, private	On-die, shared (last-level cache, LLC)
Purpose	Feed pipeline every cycle	Absorb L1 misses	Absorb L2 misses; reduce DRAM traffic

Analysis

L1 cache is split into separate instruction and data caches and sits directly on the critical path of every load/store; its extremely low latency (1–4 cycles) is essential because any delay here stalls the pipeline immediately. Its small size keeps hit latency low but also limits the hit rate for large working sets.

L2 cache acts as a victim/inclusion cache for L1 misses. It has a larger capacity (256 KB–1 MB), so it captures a large fraction of L1 misses, but its access latency (8–15 cycles) is higher because it is physically farther from the core and often uses a different (denser) SRAM array.

L3 cache (Last Level Cache) is shared across all cores on the chip. Its main role is not per-access speed but overall system throughput: by holding tens of megabytes of recently used data, it prevents a large number of requests from going all the way to DRAM, whose latency (100+ cycles) would otherwise stall multiple cores simultaneously. Because it is shared, its effective bandwidth per core drops under heavy multi-core contention, but the aggregate throughput benefit of avoiding DRAM traffic outweighs this cost.

Impact on Processor Performance

- A well-tuned hierarchy minimizes Average Memory Access Time (AMAT) using AMAT = HitTime(L1) + MissRate(L1) × [HitTime(L2) + MissRate(L2) × MissPenalty(L3/DRAM)].

- Higher hit rates at L1/L2 directly reduce pipeline stalls, improving IPC (instructions per cycle).
- L3, by reducing DRAM accesses, protects overall system bandwidth in multi-core workloads and reduces memory-controller queuing delay.
- The three-level design is a latency-bandwidth-capacity trade-off: each level is only as fast as it needs to be to keep the level above it fed, while providing enough capacity to reduce traffic to the level below.

Question 2: SRAM vs DRAM — Performance and Justification of Use

SRAM (Static RAM) and DRAM (Dynamic RAM) are the two primary semiconductor memory technologies used in the memory hierarchy. Their internal cell structures create fundamentally different latency, bandwidth, cost, and density characteristics, which decide where each is used.

Comparative Table

Parameter	SRAM	DRAM
Cell Structure	6-transistor (6T) flip-flop, no capacitor	1 transistor + 1 capacitor (1T1C)
Access Latency	Very low (~1–2 ns)	Higher (~50–100 ns)
Bandwidth	High — directly matched to core clock	Lower than SRAM; limited by row activation/precharge and refresh cycles
Throughput	Sustains near CPU-clock-rate accesses	Lower sustained throughput; interrupted by periodic refresh
Refresh Requirement	None (static — holds data as long as powered)	Needs periodic refresh (charge leaks from capacitor)
Density / Cost per bit	Low density, high cost per bit	High density, low cost per bit
Power Consumption	Higher static power (more transistors), no refresh power	Lower static power, but refresh adds dynamic power overhead
Typical Use	Cache memory (L1/L2/L3), registers	Main memory (system RAM)

Justification

SRAM is used for cache memory because caches must return data within a few CPU cycles to avoid stalling the pipeline. The 6T cell provides very fast, glitch-free read/write without needing a refresh cycle, giving SRAM the low, predictable latency that cache design demands. Its main disadvantage — large cell area and high cost per bit — is acceptable for caches because they are intentionally small (KB to a few MB).

DRAM is used for main memory because systems need gigabytes of capacity at low cost. The 1T1C cell is far smaller than SRAM's 6T cell, so DRAM achieves much higher density and lower cost per bit, at the expense of latency: the storage capacitor must be read (destructively) and rewritten, and it also leaks charge, requiring periodic refresh (typically every 64 ms). This makes DRAM roughly 20–50x slower per access than SRAM but far more economical for bulk storage.

Interpretation

- The SRAM/DRAM split reflects a classic latency-vs-capacity trade-off in the memory hierarchy: fast-small (SRAM/cache) close to the core, slower-large (DRAM/main memory) farther away.
- Using DRAM for cache would make caches too slow to be useful; using SRAM for main memory would be prohibitively expensive and power-hungry at gigabyte scale.
- This complementary placement minimizes overall AMAT while keeping system cost and power within practical limits.

Question 3: Virtual Memory, Paging, and Cache — Memory Management Mechanisms

Virtual memory, paging, and cache memory are complementary mechanisms that together create the illusion of a large, fast, and protected memory space while using a small amount of fast physical memory efficiently.

Comparative Table

Parameter	Cache Memory	Virtual Memory (with Paging)
Purpose	Bridge speed gap between CPU and main memory	Bridge capacity gap between main memory and secondary storage; provide process isolation
Unit of Transfer	Cache line/block (e.g., 64 bytes)	Page (typically 4 KB or larger)
Access Latency on Miss	Tens of cycles (fetch from next cache level/DRAM)	Millions of cycles on a page fault (fetch from disk/SSD)
Managed By	Hardware (cache controller)	OS (page table) + hardware (MMU/TLB)
Utilization Benefit	Exploits temporal/spatial locality to reduce average latency	Allows processes to use more address space than physical RAM; enables multiprogramming
Throughput Impact	High hit rate sustains near-memory-speed throughput	Excessive page faults ('thrashing') sharply reduce system throughput

Analysis

Cache memory operates purely in hardware and exploits locality of reference at a fine granularity (cache lines) to keep frequently used data close to the CPU, minimizing access latency for the common case.

Virtual memory extends this idea to a much coarser granularity (pages) and a much larger gap in speed. Paging lets the OS map a process's virtual address space onto physical frames, swapping pages to and from secondary storage as needed. This improves memory utilization by allowing more processes (or larger processes) to run than physical RAM alone would permit, and it provides isolation and protection between processes.

The Translation Lookaside Buffer (TLB) is itself a small, fast cache for page-table entries, showing how the cache concept is reused inside the virtual-memory mechanism: a TLB hit avoids a full page-table walk, keeping address translation from becoming a bottleneck.

Interpretation — Contribution to System Performance

- Cache memory minimizes latency for the CPU-to-DRAM gap; virtual memory/paging manages the DRAM-to-disk gap and enforces protection.
- Both rely on the principle of locality: caches on temporal/spatial locality of data and instructions, paging on the working-set behavior of programs over time.
- A high page-fault rate (poor locality or insufficient RAM) causes thrashing, which can degrade throughput far more severely than a high cache-miss rate, because disk access is orders of magnitude slower than DRAM access.

- Together, these mechanisms form a coherent hierarchy — TLB/cache for speed, paging/virtual memory for capacity and protection — that maximizes effective system throughput within physical memory and cost constraints.

Question 4: NAND Flash vs DRAM — Latency, Bandwidth, Throughput, and Suitability

NAND Flash and DRAM occupy different levels of the memory-storage hierarchy. DRAM is volatile main memory optimized for speed, while NAND Flash is non-volatile secondary storage optimized for density, persistence, and cost.

Comparative Table

Parameter	DRAM	NAND Flash
Volatility	Volatile (loses data on power-off)	Non-volatile (retains data without power)
Access Latency	~50–100 ns	~25–100 microseconds (read); write/erase far slower
Bandwidth	High (tens of GB/s per channel)	Lower than DRAM, though modern NVMe SSDs achieve several GB/s via parallel channels
Throughput	High random and sequential throughput	High sequential throughput; random writes limited by erase-before-write and wear leveling
Write Behavior	Direct overwrite, symmetric read/write speed	Must erase a block before rewrite; write amplification
Endurance	Effectively unlimited read/write cycles	Limited program/erase (P/E) cycles (thousands to ~100,000 depending on type)
Cost per Bit	Higher	Lower — enables large-capacity, low-cost storage
Typical Role	Main (working) memory	Secondary storage — SSDs, embedded storage

Analysis and Suitability

DRAM's nanosecond-level latency and symmetric high-speed read/write make it the only practical choice for main memory, where the CPU needs to fetch instructions and data continuously at near-clock-speed rates. Its volatility is an acceptable trade-off because main memory content is expected to be transient and reloaded/refreshed as needed.

NAND Flash trades speed for persistence and density. Its microsecond-level access latency is far too slow to sit between the CPU and cache, but this is irrelevant for its role as secondary storage, where the comparison point is not DRAM but rotating hard disks (which are even slower, at millisecond latency). NAND Flash's block-erase requirement and limited P/E-cycle endurance are managed by SSD controllers using wear leveling and over-provisioning.

Suitability in the Hierarchy

- DRAM: main memory tier — must be fast and directly addressable by the CPU; volatility is acceptable given its role.
- NAND Flash: secondary storage tier (SSD) — bridges the gap between DRAM and slower archival storage, offering non-volatility, high density, and much lower cost per bit than DRAM.

- Emerging use: NAND Flash (and NVMe protocols) increasingly serve as an extension of memory (e.g., swap, memory-mapped storage) in systems with large working sets, though at higher latency than true main memory.

Question 5: MRAM vs RRAM — Access Time, Throughput, Energy Efficiency, and Endurance

MRAM (Magnetoresistive RAM) and RRAM (Resistive RAM) are emerging non-volatile memory (NVM) technologies positioned to close the gap between fast-but-volatile SRAM/DRAM and dense-but-slow NAND Flash. Both aim to combine speed, non-volatility, and low power for future hierarchical memory systems.

Comparative Table

Parameter	MRAM	RRAM (ReRAM)
Storage Mechanism	Magnetic tunnel junction (MTJ) — resistance set by relative magnetization direction	Resistive switching in a metal-oxide layer via formation/rupture of a conductive filament
Access Time	Very fast (~2–20 ns), close to SRAM/DRAM	Fast (~10–50 ns), slightly behind MRAM in mature designs
Throughput	High; suitable for cache/DRAM-replacement roles	High; competitive, though variability between cells can limit sustained throughput
Energy Efficiency	Low read energy; write energy moderate (STT-MRAM improves this over toggle-MRAM)	Very low read/write energy per bit; strong candidate for ultra-low-power systems
Endurance	Very high (10^12–10^15 cycles) — near DRAM-like endurance	Moderate (10^6–10^12 cycles depending on material) — generally lower than MRAM
Non-Volatility	Yes	Yes
Density/Scalability	Moderate; MTJ stack limits extreme scaling	High; simple two-terminal structure scales well and enables 3D crossbar arrays
Maturity	Commercially available (STT-MRAM in embedded/niche products)	Still emerging; used in some embedded and neuromorphic applications

Analysis

MRAM offers the best balance of speed and endurance among emerging NVMs, making it attractive as a potential replacement for SRAM in cache or as a persistent, fast working memory. Its main limitations are relatively higher write energy compared to RRAM and a cell structure that is harder to scale to extremely small feature sizes.

RRAM offers excellent scalability and very low switching energy, and its simple two-terminal structure allows dense 3D crossbar integration, giving it strong potential for high-capacity, low-power storage-class memory. However, cycle-to-cycle and device-to-device resistance variability can affect reliability and endurance compared to MRAM.

Potential Applications and Interpretation

- MRAM: last-level cache replacement, persistent registers, and non-volatile working memory in systems requiring instant-on capability and radiation tolerance (e.g., aerospace, industrial control).
- RRAM: storage-class memory (SCM) bridging DRAM and NAND Flash, high-density embedded non-volatile memory, and neuromorphic/in-memory computing due to its analog resistance states.

- Both technologies point toward future hierarchical memory systems where the traditional volatile/non-volatile boundary blurs — enabling 'persistent memory' tiers that combine near-DRAM speed with Flash-like non-volatility, reducing energy spent on refresh and data movement.
- Trade-off summary: MRAM currently wins on endurance and maturity; RRAM currently wins on scalability and switching energy — the right choice depends on whether the target application prioritizes lifetime (MRAM) or density/power (RRAM).